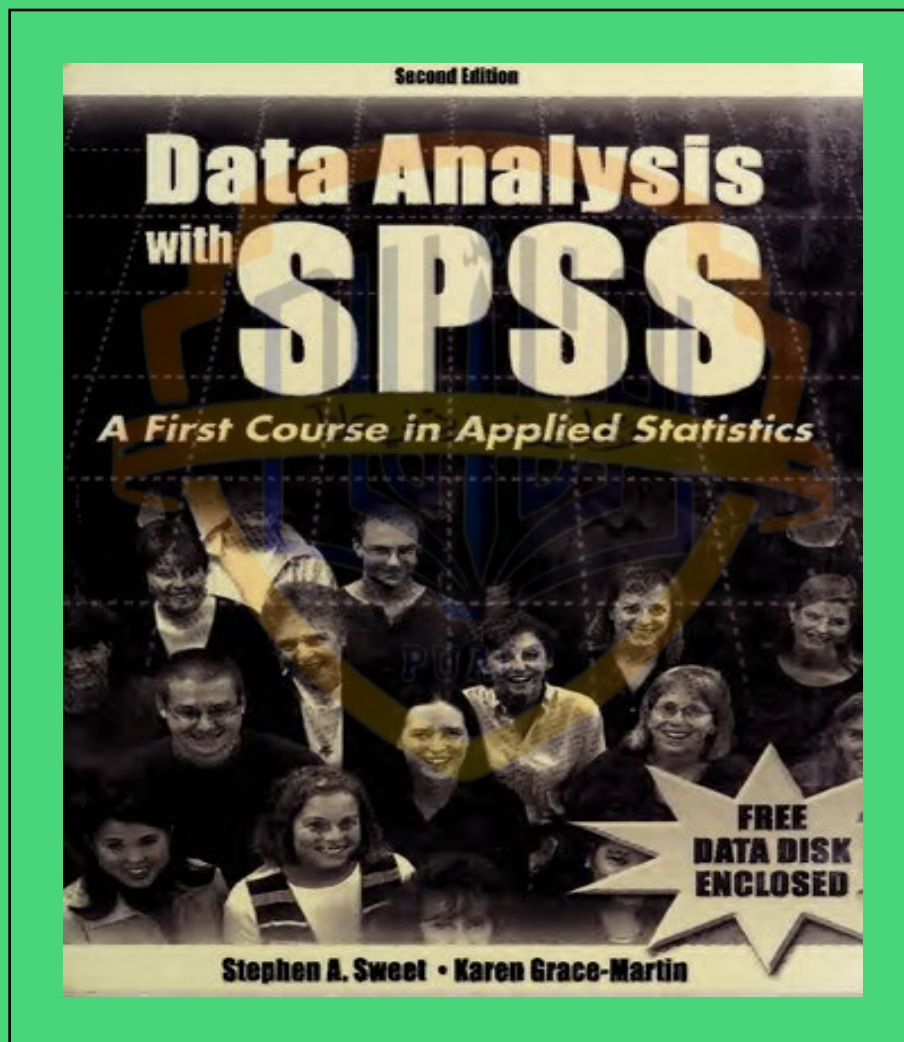


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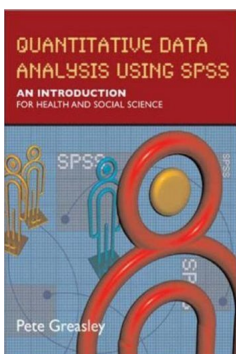
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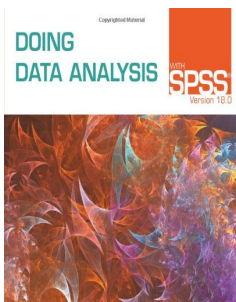
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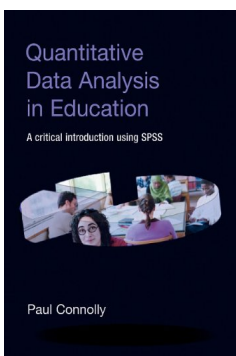
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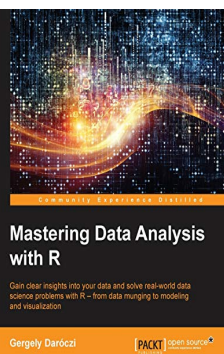


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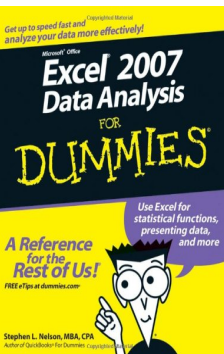


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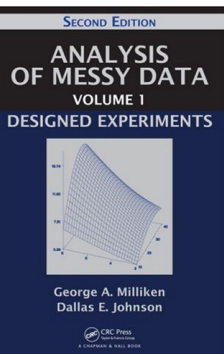
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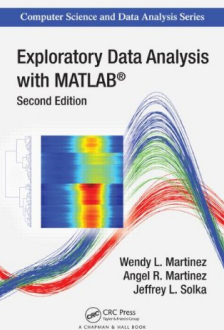
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Second Edition

Stephen A. Sweet
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Cornell University

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Preface

This book is written for anyone pursuing a career in the social sciences, as well as those needing to learn the essentials of statistical methods. We have written less about statistics than about how to take real world observations, expose these observations to statistical analysis, and understand how to systematically study social relationships. As such, this book can offer interest to students in psychology, sociology, and education, as well as those in other disciplines relying on statistical methods. We provide an active hands-on approach to learning the skills of data analysis, using the Statistical Package for the Social Sciences (SPSS) and some of the most current data on social behavior available.

We believe that statistics are best learned in *context*, as they are applied to analyzing real world observations. Our hopes are that readers will realize what we learned long ago, that statistical analysis (contrary to popular belief) can be fun. This assertion will strike many readers as surprising and counterintuitive. We think *one of the reasons* traditional statistics courses are not well received by most students *simply stems from focusing too much* on the ways statistics are *generated*, and not enough on *how* statistics are *interpreted*. In contrast to the methods of calculating statistics, the interpretation of statistics involves applying *the appropriate* statistical methods and then analyzing the *results* to gain an understanding of people and society.

We also believe that statistics become interesting when they are *applied*. We do not view statistics as a means of testing *mathematical skills*; rather, our interests concern how statistics can be used to gain *understandings of the social world*. *Do social policies work the way they were intended? Does a social program make a difference? In what ways do different groups of people vary from one another?* In this book, we try to chart effective pathways to answering these types of research questions using real data in the same manner that professional social scientists would.

To develop skills, we introduce *key concepts* and statistical approaches commonly used by social scientists, and then conclude *chapters with hands-on exercises*. These exercises draw from two data sets, rich with information *concerning social behavior*. One of these data sets contains hundreds of variables about *social behavior in the United States*, focusing on differences between states. Among other topics, *these data enable analysis of education, criminality, welfare use, health care, taxation, births, mortality, and environmental well-being*. The other data set contains nearly 100 variables from the 1998 General Social Survey conducted by the Roper Center for Public Opinion. These data enable analysis of variation in individual experiences and attitudes relating to education, religion, politics, sexuality, race, and taxation. These data sets are used to build analytic skills in a step by step manner, from the initial stages of forming research questions and collecting data through advanced multivariate methods and report writing. As these two large data sets offer abundant information on places and people, they open a vista of possibilities for independent research activities.

Do I Have What It Takes and What Will I Accomplish?

There is something about statistics which seems to spark fear. Computers do likewise. Put the two together and dread may result. This is unfortunate and we believe unnecessary. Some readers (possibly you?) may have a few concerns at this point. Some may be wondering if they need to know how to use a computer to get through the exercises in this book. The short answer is yes, but there is no time like the present to learn! Prior computer experience would be

of help, but a basic understanding of how to navigate in a Windows environment will be sufficient to work through the exercises in this book.

Some readers may be concerned that they lack the requisite skills in statistics or math. This text is designed to help you become refined in analyzing quantitative data; therefore existing skills in statistics and math can certainly be a help. However, this book is designed to teach basic statistical inference. Therefore, no prior statistics courses and only modest mathematical capabilities are needed to perform the exercises in this text.

Finally, some readers may wonder why they should bother with this book if they do not plan to become statisticians or social science researchers. On one level, there may be direct benefits. All of us live in a world in which numbers are used to explain our experiences. Even if one becomes an artist, the ability to read statistics critically facilitates making important decisions such as where to locate one's family, what political candidates to vote for, and what drugs should be avoided. In all likelihood, SPSS will not be used to make these decisions, but skills in quantitative data analysis can contribute to making the best informed decision. At a deeper level, understanding how data are analyzed can help all readers develop a critical perspective on how to evaluate arguments based upon quantitative data. Indeed, the real aim of this book is not so much about teaching SPSS. It is to cultivate the skills needed to be a critical consumer of information and to whet the appetite for empirical inquiry.

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Many people contributed to this book at various stages of its development. Scott Morgan and Kathleen O'Leary Morgan were exceedingly generous in allowing us to reproduce the data presented in their outstanding books: *State Rankings 2000*, *Health State Rankings 2000*, and *Crime State Rankings 2000*. The Roper Center for Public Opinion Research allowed us to select and include variables from the 1998 General Social Surveys, data from one of the very best public opinion polls available.

The Cornell Office of Statistical Consulting offered great help. The director, Ed Frongillo, provided excellent suggestions for refining our writing, as well as offered great support for the development of this book. Karen's colleagues Francoise Vermeulen and Yun Wang bore the burden of extra consulting while she wrote.

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We owe a deep debt of gratitude to Catherine Soria, who spent many hours arranging data, proofreading, and editing, and re-editing. Also thanks to Alex Pearson and Pavel Krivitsky, who helped organize the data sets. Jeff Lasser, our editor, at Allyn and Bacon, offered excellent advice throughout the project.

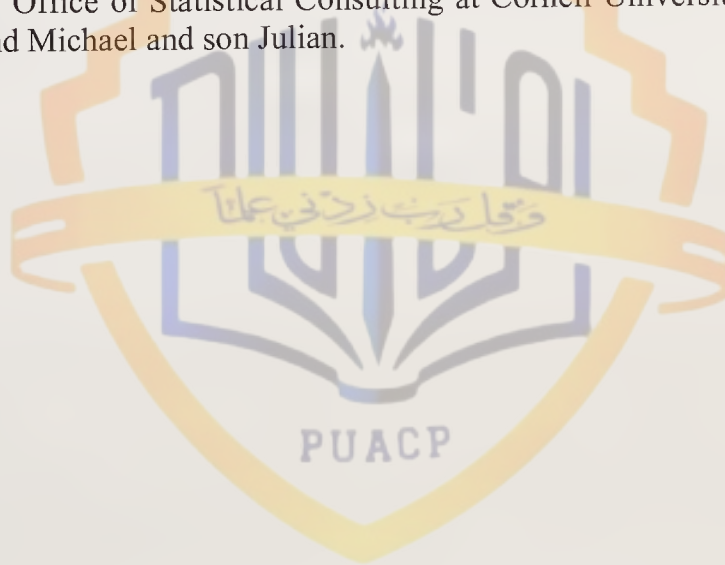
Most of all, we thank Karen's husband Michael, and Stephen's wife Jai, who gave unwavering love and support and spent many hours babysitting so we could write.

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Chapter 1

Key Concepts in Social Science Research

Overview

We begin our introduction to data analysis with the major concepts social scientists use to develop research questions, understand data, and test relationships. We do not detail how to use SPSS in this chapter. Instead, we build a foundation for using SPSS in a manner consistent with the practices of professional social science. This includes adhering to empiricism, the careful consideration of causal relationships, forming testable hypotheses, and developing valid and reliable measures. All of these concerns are relevant to analyzing the data provided in this book and successfully launching an independent research project.

Empiricism and Social Science Research

Most social scientists spend their professional lives studying the human condition. Sometimes they study the occurrence of conditions or behaviors, such as the degree to which families experience poverty, the frequency of criminal behaviors, or the proportion of the population attaining educational degrees. Gathering this information over time allows researchers to identify trends and predict future patterns. In other circumstances, they test relationships—the ways in which one set of conditions or behaviors potentially affects another. For example, criminologists identify the factors that predict crime and assess the outcomes of treatment programs. Poverty researchers assess the impact of programs or policies on social relations, especially among poor people and their communities. Similarly, one of the key concerns for education researchers is identifying factors that contribute to success in attaining high school and college degrees. These efforts require rigorous methods of observation and analysis, which are, in a word, **empirical**. The doctrine of empiricism asserts that the combination of logic and observation builds understanding.

The empirical nature of scientific inquiry created the need for statistical analysis—methods for understanding and linking observations with one another. Statistics developed because scientists agreed that it was not enough to think abstractly about how the physical and social worlds operate or to take everyday beliefs as true, but to expose beliefs to systematic measurement and rigorous analysis. Statistics is one means of critically studying how well beliefs correspond to real world circumstances.

Social scientists face considerable challenges as they strive to be empiricists. They should hold a steadfast commitment to logic and observation, and not advance politics or self-interest. It is natural to want to find results that support personal beliefs or values. However, as empiricists, scientists strive to carefully report the picture of the social world that is revealed in analyses, no matter what the implications. For instance, suppose we believed that the death penalty was an effective deterrent to crime and began a series of analyses to test our beliefs. If our results concluded the opposite, we would be obligated to report it. As the sociologist Max Weber explained (Gerth and Mills 1967), social scientists are obligated to be **value neutral** in their research. Empiricists set aside their egos and politics when they begin data analysis.

Data

This book offers an introduction to social science research by applying statistical applications to the study of two data sets, one documenting the characteristics of people, the other documenting characteristics of places. We depart from traditional methods for teaching statistics by focusing less on statistical calculations and more on how to understand what statistics reveal. We try to create a context where newcomers can apply rigorous statistical methods to their own empirical studies. This research-oriented approach teaches the skills needed for data analysis in an applied fashion. Toward this end the data sets included with this book are structured to enable newcomers to immediately use statistics and put them to practical use.

One data set, the **General Social Survey (GSS98) data**, contains information on people. These data are based on telephone interviews of men and women who responded to a series of questions concerning their attitudes, behaviors, and beliefs. Questions include, among other things, demographic information such as ethnicity, income, age, and education. Other questions tap into beliefs and behaviors, such as the degree to which individuals hold prejudicial attitudes and the amount of television they watch. Because these measures were developed through years of testing and because the respondents are drawn from a nationally representative sample of households, these data are among the very best available. The other data set, the **STATES data**, contains information on states and the types of people residing there. This information includes poverty rates, alcohol sales, crime rates, and a variety of other information about life within each state. The sources of these data include the U.S. Census, National Education Association, Department of Labor, the Internal Revenue Service, and the U.S. Department of Health and Human Services. These are among the most reliable sources of information on the social condition of the nation.

These two files are **data sets**, organized collections of information. This book addresses how to approach a data set in a systematic manner and how to launch a research project involving statistical analysis. Any research project requires collecting and understanding **data**, pieces of organized information about elements in a population (a **datum** is a single piece of information). We have taken care of the first task and collected some of the best quality data available to date on people and places. In fact, these two data sets are comparable, and in many

instances identical, to the types of data that professional social scientists use to study social life. They are also the types of data policy makers use to inform legislative decisions. Our hopes are that these data will also help novice researchers understand statistics and how data analysis is applied in social science research. Those already beginning to think about research projects may find it useful to skim the appendices at the end of this book, where the data sets are described in greater detail, and to see if these can provide information to correspond with existing research interests. For others, this information may help kindle ideas and spark the imagination for a wide variety of potential projects.

Data sets are structured to link cases with variables. A **case** is the individual unit being studied. Each case has one datum, or **observation**, for each variable. For the GSS98 data, each case is an individual person. In the STATES data, each case is the individual state, such as New Hampshire. **Variables** contain information about each case and are structured in a way that makes one case comparable to another. For instance, in the GSS98 data, all of the respondents were asked their age, and as a result, one could compare any one respondent's age with that of another. The STATES data have a variable indicating crime rates, and therefore states can be ranked in terms of how much crime occurs within their geographic areas. When data are collected in a systematic manner, researchers can analyze the patterns between variables. For instance, having information on crime rates and whether the death penalty exists in the fifty states allows a researcher to analyze the relationship between these variables.

Although these two data sets are available in this book, and can help expedite many research projects, there are many other sources of data available over the Internet. As researchers consider using these resources, they need to consider their strengths and weaknesses, as data quality can vary markedly. Empiricists understand that the goals in research should be to gather and construct data that are of the highest quality. It is not enough to have any data. Data should be well suited to the issues of interest. Selecting data sets and variables is challenging, as the quality of these choices will influence the acceptability of all subsequent research activities. Even with the application of the most sophisticated statistical methods, the acronym GIGO holds true: garbage in, garbage out.

When selecting variables for study, the scientist strives to find **indicators** that best capture abstract concepts. For example, studies of criminality require using some indication of the occurrence of crime. One possibility is to use data from surveys assessing victimization, asking people if they have ever been crime victims. As many crimes are never reported to the police, this indicator will likely offer different information than arrest records or police reports. Studies of educational achievement could involve examining grade point averages, graduation rates, or scores on standardized tests.

As researchers select and construct indicators, they strive to make them as reliable and valid as possible. **Reliability** refers to the degree to which an indicator is a consistent measuring device. It is no accident that carpenters use metal tape measures instead of strings and navigators use compasses instead of star gazing. Strings and stars work, but not nearly with the consistency offered by the other methods. The same concern extends to social research, and researchers strive to select consistent tools for measuring concepts such as alienation, alcohol consumption, and depression. Consider the implications of assessing educational achievement by asking students "how well do you do in school?" Unless all students use the same standards for self-assessment, the results for different students might not reflect their achievements, but rather the different standards they hold for what counts as achievement. In such circumstances the reliability of this measure becomes questionable. For this reason, SAT scores, or other standardized test scores,

tend to be favored. Similarly, one could conceivably measure school funding by asking individual property owners how much they pay in school taxes. As this indicator would rely on the homeowner's memory, the degree to which this indicator provides accurate assessments would certainly be inferior to the types of data one could collect at a town clerk's office.

Validity refers to the indicator measuring what it is supposed to measure. No one uses a steel ruler to measure temperature, simply because it was never designed for that purpose. Although this concern seems straightforward enough, novice researchers sometimes select indicators that are loosely related to the abstract concept they are really interested in studying. For example, IQ scores are sometimes erroneously used to measure educational achievement. Poverty is inappropriately used as a measure of criminality. The error in both examples is confusing a preconceived relationship with the variables intended to be measured.

To recap, social science is built on an adherence to empiricism, and for many studies this requires the use of statistics. Statistics are used to measure patterns of individual variables and relationships between variables. As social scientists develop studies, they set aside their personal agendas in favor of a dispassioned analysis of data, reporting results accurately and without bias. As they develop their research projects, they select data sets that will offer the most accurate indicators for the concepts they are interested in studying, seeking variables that are reliable (consistent) and valid (appropriate).

Developing Research Questions

We suspect that anyone who was asked why teenagers drop out of school could offer a number of plausible ideas. Some likely causes are overcrowding of classes, insufficient funding for schools, a loss of respect for authority, drugs, boring textbooks, influence of gangs, racial discrimination, and so on. Everybody has theories that they use to explain patterns of social behavior. Indeed, symbolic interactionists have long argued that without constant theory making, people would have great difficulty creating meaningful social encounters (Blumer 1969). What distinguishes social science researchers from armchair theorists is the degree of formality they use in constructing their research questions and the rigor with which they expose their theories to systematic study (Sweet 2001).

The initial step in any research project is to identify the **research question**. Developing a research question involves taking an issue, often something most people take for granted, and posing it in the form of a question. Developing a research question puts a researcher on a pathway of knowledge. Questions highlight a void, or an absence of understanding. The job of scientists is to fill this void and to resolve or explain something that, until the question was raised, was left unanswered. The research question should drive social science inquiry, not the desire to establish a particular finding as "true."

Research questions are actually very liberating for the scientist. There is a very big difference between starting a study with the intent of showing, for instance, that capital punishment is beneficial to society, and starting a study with a question "Is capital punishment an effective deterrent to crime?" The first framework leads a researcher to selectively gather information to support his or her existing stance. In contrast, by making the effects of the death penalty the question, the researcher creates a condition that forces him or her to think critically about the types of information to gather and the relationships to test. As a result, the researcher becomes unencumbered from the desire to report only the information that supports a predetermined conclusion, and creates a situation where curiosity drives analysis.

Social scientists pose questions so that unexpected results can emerge and be seriously considered. “Is there racial discrimination in the education system?” is a much better way to pose a research project than “I want to show that there is discrimination in the education system.” When the research project is posed as a question it assists the researchers’ efforts to identify a clear analytic strategy, and to think critically about the types of information needed to test relationships. In the process of developing a research question, novice researchers (as well as seasoned scientists) often realize that their initial ideas are too broad to be managed in one study. They can then parse out their main question into manageable subquestions, and select those questions that they are most interested in immediately pursuing. For example, some interesting subquestions about discrimination in education are:

Is there an equitable representation of minority groups within the curriculum?

Is there proportionate employment of minority members in school systems?

To what extent do graduation rates vary among racial groups?

Do minority school districts receive comparable funding to those in predominantly white school districts?

As a researcher works toward developing a study, it is important to perform a **literature review** in the early phases of the project. The literature review is an overview of past research on topics relating to the research question. The best sources for literature reviews will be scholarly journals such as *The American Journal of Sociology*, *Child Development*, *Journal of Marriage and the Family*, and *Journal of Health and Social Behavior*, to name a few. Most articles in scholarly journals have a detailed description of how data were collected and offer extensive references to other related studies. These sources are available through key word searches of computer databases located in university libraries.

The literature review does three things. First, it helps the researcher appreciate the knowledge that has been gained on a subject to date. Second, the literature review informs the researcher of the methods and analyses other scholars used to answer similar research questions. Finally, it helps researchers contribute to the cumulative building of knowledge by integrating current research activities with previous literature. When researchers examine the literature and identify questions that still need to be answered or methods that have yet to be tried, they position themselves to perform new and vital work.

One of the best strategies for novice researchers is to formalize a tentative research question. This should be made into a single question, such as “to what degree does X influence Y?” or “does X lead to Y?” or “what factors contribute to Y?” Once the question is formalized (ideally in writing), the researcher should do a literature review, surveying what others have written on the subject. Although this initially may seem like adding work to the data analysis project, in the long run it will save effort and create a more meaningful and higher quality study. Remember that existing studies offer examples of ways to develop research projects, as well as highlight considerations germane to the project.

Theory and Hypothesis

Finding answers to research questions involves developing theories and hypotheses. According to traditional scientific canons, a **theory** is an unproven explanation of facts or phenomena. Theories should generate **hypotheses**, falsifiable statements concerning expected relationships. Hypotheses should be structured in anticipation that they will be supported or refuted when exposed to some type of test. They are also structured in such a way that when the test is performed, unexpected results can potentially occur and the preexisting beliefs will not necessarily be supported. In a word, hypotheses should be constructed so that they are falsifiable.

Many social scientists use the **deductive approach**, using a theory to develop a number of hypotheses to be tested before they begin their data analysis. For example, we might theorize that inequities in school funding lead minority groups to perform, on average, worse in school than white students. The theory helps us formulate some hypotheses. Some reasonable hypotheses are:

- H1: States with higher proportions of African Americans and Hispanics will have lower school funding.
- H2: States with higher levels of school funding will have higher graduation rates.
- H3: States with higher levels of school funding will have students performing better on standardized tests.

Note that each hypothesis is written so that it is testable and can be refuted. Falsifiability is important because it allows unexpected findings to emerge and our expectations to be challenged in the light of critical observation.

Another illustration can be drawn from the theory that the death penalty is effective at deterring crime. This theory produces the following testable hypotheses:

- H1: States that have the death penalty will have lower property crime rates.
- H2: States that have the death penalty will have lower murder rates.
- H3: States that have the death penalty will have fewer criminals reenter jail after being released (have lower “recidivism” rates).

It is important to acknowledge that not all scientists follow the deductive approach when analyzing data, and that some social scientists follow an **inductive approach** termed **grounded theory** (Glaser and Straus 1967). Grounded theory, like traditional deductive social research, relies upon hypotheses, indicators, and well-formed research questions. However, unlike the traditional deductive approach, tentative theories are based on data and researchers continually refine the theories to concur with analyses. The research project ends when the researcher understands the relationships in the data and develops a theory that explains these relationships. **Exploratory data analysis** is similar in intent (Hoaglin, Mosteller, and Tukey 1983). Advocates suggest that prior to running advanced statistical procedures for a critical test of a hypothesis,

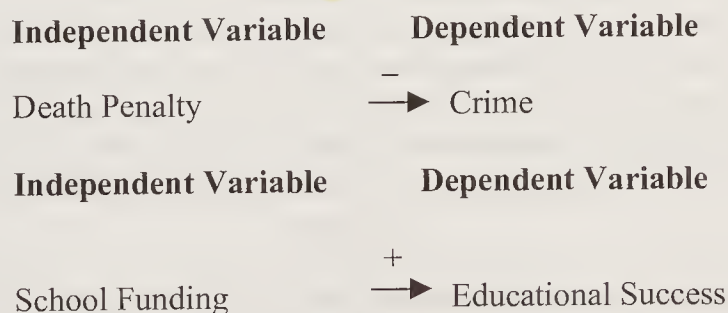
researchers should get involved with the data and explore relationships using highly visual depictions of relationships.

Which is better—the more traditional deductive approach or the inductive approach? One is not necessarily better, nor are they as independent as is often supposed. We think that novice researchers can benefit from using both approaches in their own data analyses. After developing a research question, the next step is to briefly explain what they think the answer will be before beginning analysis to create a rudimentary theory. Then the theory can be parsed out into testable hypotheses, made in consideration of the available data. This step-by-step process is reflective of the deductive approach. However, once the novice researcher has developed or accessed a data set, he or she can also do some exploratory work, testing relationships with a bit more freedom, exploring the data and relationships within the data set. This exploration is not the end in itself; rather, it can open the mind to different types of relationships and to theories that might explain these relationships.

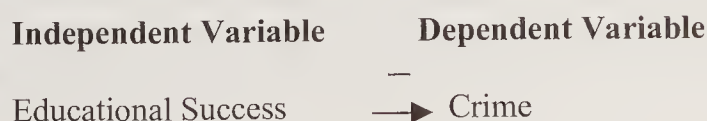
Relationships and Causality

Most social science research is about **causal relationships**, how a particular set of conditions leads to predictable outcomes. Seldom will such relationships be **deterministic**, producing inevitable results. More often, social science documents **probabilistic relationships**, in which factors increase tendencies toward particular outcomes. For example, consider the conclusion that education increases income. If this were a deterministic relationship, everyone with a college degree would have higher incomes than everyone with a high school degree. This, of course, is not the case. But it is the case that those with college degrees *tend* to earn more than those with high school degrees.

How does one determine if a relationship is causal? Answering this question requires identifying the relationships between independent and dependent variables. **Independent variables** are hypothesized to cause changes in **dependent variables**. We have already discussed a number of hypothesized causal relationships, such as the death penalty deterring crime and school funding increasing educational success. Diagrams of these relationships would be:



The very same indicators can switch from being independent variables to dependent variables, depending upon the research question being addressed. For example, increased educational success could potentially lead to lower crime rates:



Understanding the differences between independent and dependent variables is crucial. To assert a causal relationship is to claim that changes in the independent variable create changes in the dependent variable. In practice, researchers should only assert that one factor causes changes in another variable when they can reasonably satisfy the following three criteria:

1. *Association*: there must be a relationship between the independent and dependent variables.
2. *Time order*: the change in the independent variable must precede the change in the dependent variable.
3. *Nonspuriousness*: the effect of a third unmeasured “spurious” factor does not produce the relationship between two variables.

Association

When two variables are associated with one another, both variables correspond in predictable ways. **Associations** can either be **positive** (an increase in one variable corresponds with an increase in the other) or **negative** (an increase in one variable corresponds with a decrease in the other variable). For instance, the statement “people with higher degrees tend to have higher incomes” is a statement of positive association. “People who achieve higher levels of education tend to be less prejudiced” is a statement of negative association. Note that neither statement claims a causal relationship, only that changes in the variables correspond with one another.

It is important to note that many associations are not causal. For instance red cars are in more accidents than white cars. It isn't the red cars that are causing the accidents, it is the tendency of aggressive drivers to purchase these red cars and get into accidents (see below).

Time Order

When a change in one variable causes another variable change, logically, the independent variable needs to change before the dependent variable. For instance, the statement “instituting the death penalty will decrease crime” is a statement implying time order. Making this assertion requires having measurements of criminality at two points in time, prior to the institutionalization of the death penalty and after the death penalty has been enacted. The issue of time order seems straightforward enough. However, establishing time order can be difficult because most data are only collected at one point in time. As such, asserting the existence of a causal relationship has to be done with care. Also, as time often introduces changes in other variables, even when this condition is satisfied, causality may still not be the case (see below).

Nonspuriousness

A **spurious relationship** exists when two variables appear to be causally related, but that relationship is caused by the presence of a third unmeasured variable. In spurious relationships, if the third unmeasured variable is taken into account, the relationship between the initial two variables disappears. A classic example of a spurious relationship is that of ice cream consumption causing drowning deaths. There is an association between ice cream consumption and drowning (more people drown during times when a lot of ice cream is being consumed). The time order issue can also be satisfied (increases in ice cream sales precede increases in drowning deaths). Of course the unmeasured factor in this relationship is temperature. More

people swim in the summer and more ice cream is consumed in the summer. It is in fact the warm weather operating as a spurious factor, causing both variables to increase.

Spuriousness is difficult to rule out, simply because it is impossible to account for every variable that could produce the relationship. In actual practice, good research involves measuring the factors that are theoretically likely to produce a spurious relationship. For example, even though sunspots *might* have an effect on the behavior a researcher is studying, there is no theoretical reason for this assumption, so there is no need to account for it in the analysis. However, because factors such as income and education may have a profound impact on many aspects of social life, controlling for these factors in statistical analyses is generally a good idea.

The central concern we highlight here is that researchers should be careful before they conclude that they have shown causal relationships. All three of the above criteria must be satisfied. Experimental studies, in which people are randomized to two or more groups, are valuable because they ensure time order and **nonspuriousness**. However, because experimental studies can rarely be done, the social sciences primarily rely on observational studies. When writing and discussing findings, therefore, always be careful about using the word “cause.”

Summary

Social science research hinges on the careful construction of research questions, the pursuit of the highest quality data, and the best analytic strategies. It requires researchers to think critically about their beliefs and assumptions about the ways the social world operates. This involves creating theories and distilling them into testable hypotheses. When hypotheses are constructed, data need to be gathered and organized in a manner that will enable statistical tests. Even after statistical analyses are performed, researchers have to reflect on their methodology and assess the degree to which any relationships observed can be attributed to causal processes, or if there are other potential explanations for any patterns observed in the analysis.

Key Terms

Association

Case

Causal relationship

Data

Datum

Data set

Deductive approach

Dependent variable

Deterministic relationship

Empirical

Exploratory data analysis

Grounded theory

Hypothesis

Independent variable

Indicators

Inductive approach

Literature review

Negative association

Observation

Positive association

Probabilistic relationship

Reliability

Research question

Spurious relationship

Theory

Time order

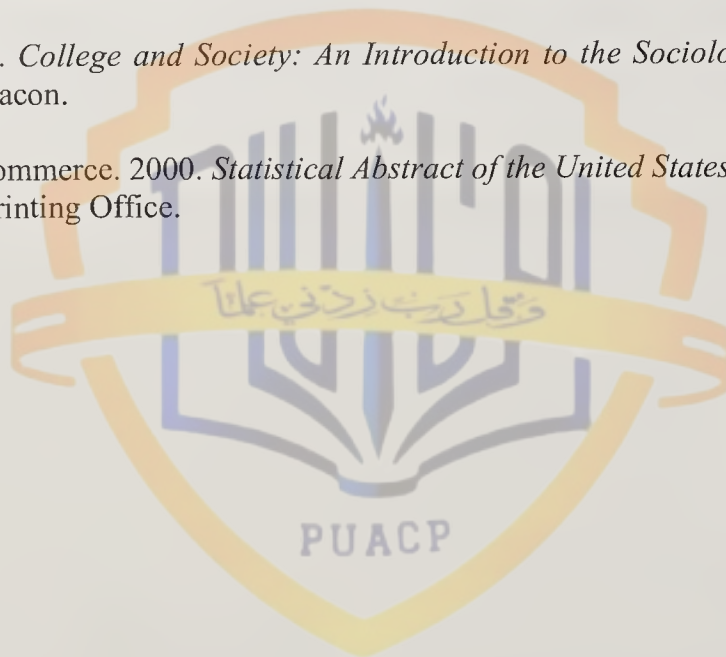
Validity

Value neutral

Variable

References and Further Reading

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Chapter 1 Exercises

Name _____

Date _____

1. Identify the independent and dependent variables in the following research projects:

- A. A study seeks to find out if listening to heavy metal music causes teenagers to become more violent than their peers who do not listen to heavy metal music.

Independent Variable_____
Dependent Variable

- B. A group of researchers is interested in examining the effects of long-term poverty. They do this by studying subjects' physiological health and attachment to the workforce.

Independent Variable(s)_____
Dependent Variable(s)

- C. A study finds that self-esteem increases as a consequence of receiving good grades on examinations.

Independent Variable_____
Dependent Variable

2. Indicators are used to measure abstract concepts. List some indicators that might prove to be reliable and valid in measuring the following concepts:

A. Economic Prosperity

E.g., Gross National Product

B. Family Violence

E.g., Admissions to Battered Women's Shelters



3. Using Appendix 1, identify the variable labels and variable names from the STATES data that might be good indicators for the following concepts:

A. Educational Attainment

Variable Name

Variable Label

Variable Name

Variable Label

Variable Name

Variable Label

B. Health

Variable Name

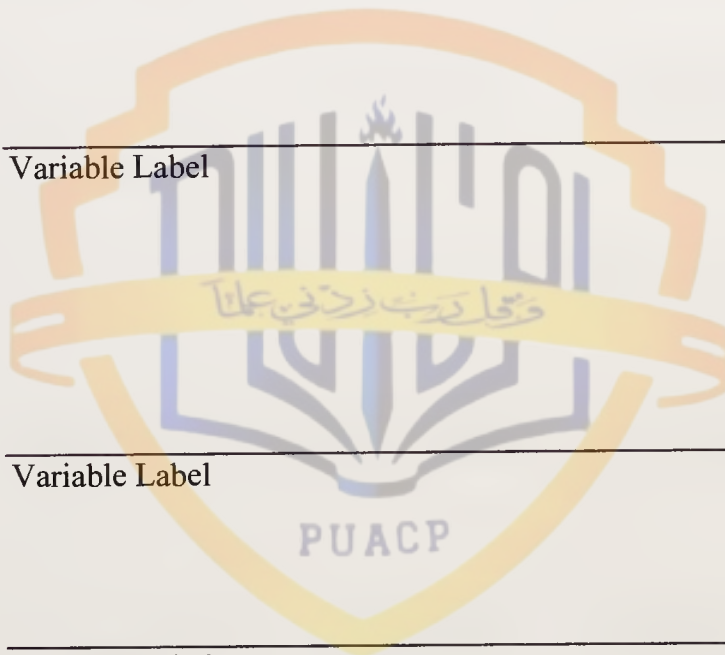
Variable Label

Variable Name

Variable Label

Variable Name

Variable Label



4. Using Appendix 2, identify the variable labels and variable names from the GSS98 data that might be good indicators for the following concepts:

A. Religious Commitment

Variable Name	Variable Label
---------------	----------------

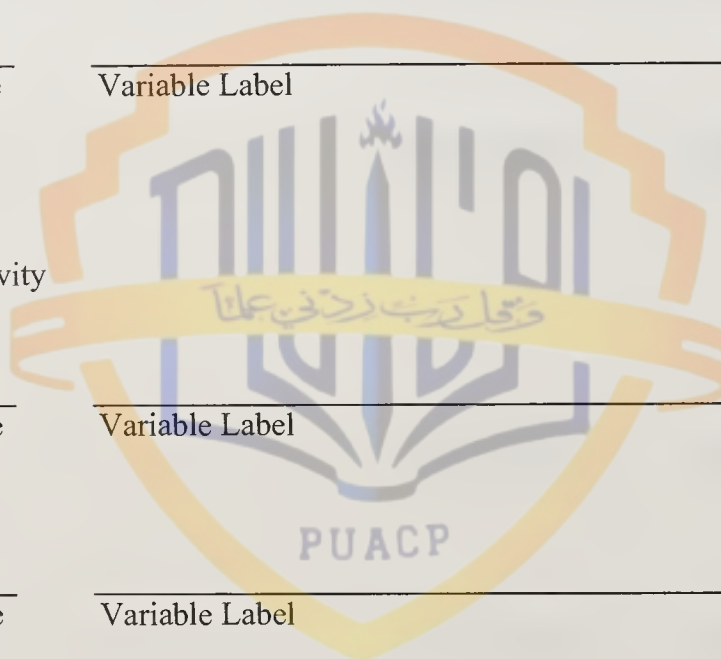
B. Prejudice

Variable Name	Variable Label
---------------	----------------

C. Sexual Activity

Variable Name	Variable Label
---------------	----------------

Variable Name	Variable Label
---------------	----------------



5. Can you identify a potential spurious factor that may call the following findings into question? Explain.

Researchers have found a consistent relationship between schools and crime. More crimes occur in neighborhoods surrounding high schools and junior high schools than in neighborhoods far away from schools. The researchers conclude that schools cause crime and believe that this relationship may have something to do with teachers not teaching students the appropriate lessons in the classroom.



6. The U.S. Department of Justice has found that of the children killed by their parents, 55% of the murders were performed by the child's mother and 45% of the murders were performed by the child's father. A researcher uses these data to support his contention that women are more violent than men. On the basis of these data, do you find this argument compelling? Why or why not?

7. A social commentator argues that the welfare programs introduced in the mid 1960s have caused an unparalleled expansion of poverty in the United States. Based on the following data from the *Statistical Abstract of the United States 2000*, would you agree or disagree with this causal statement? Explain.

<u>Year</u>	<u>Percent Below Poverty Level</u>
1960	22%
1965	15%
1970	13%
1975	12%
1980	13%
1985	14%
1990	14%
1995	14%
1998	13%

8. Select one of the following questions:

What causes crime?

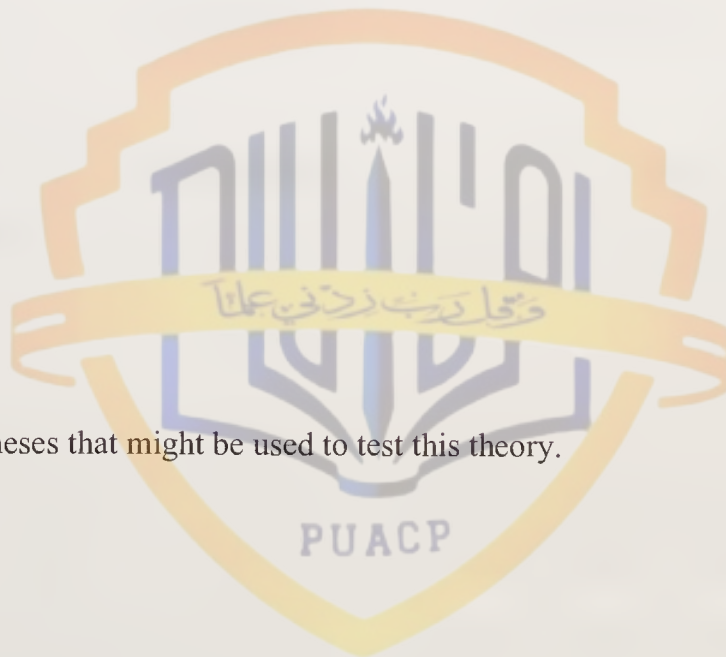
What influences educational success?

Who is most likely to be religious?

Who is most likely to be politically liberal?

A. Develop a brief theory explaining what you think is a likely explanation.

B. List a few hypotheses that might be used to test this theory.



- C. Identify any variables (using the Appendixes as the end of the book) that might be used to test the theory.

Variable Name

Variable Label

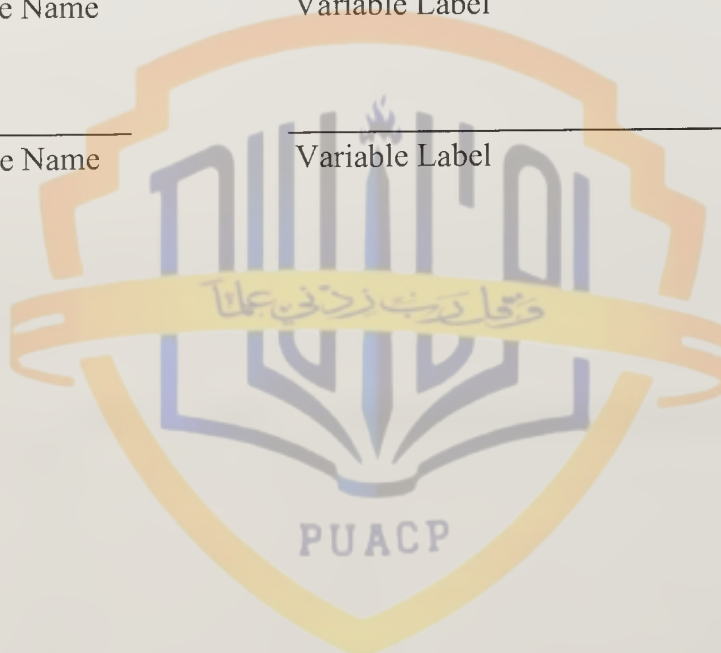
Variable Name

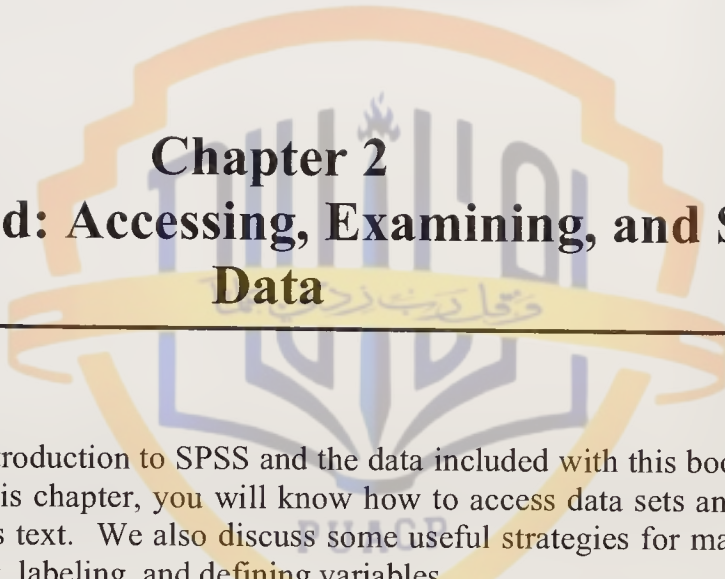
Variable Label

Variable Name

Variable Label

Variable Name

Variable Label



Chapter 2

Getting Started: Accessing, Examining, and Saving Data

Overview

This chapter is an introduction to SPSS and the data included with this book. When you complete the exercises in this chapter, you will know how to access data sets and evaluate the indicators included with this text. We also discuss some useful strategies for making data sets accessible, including naming, labeling, and defining variables.

We suggest that you work through the chapters with the SPSS program running on your computer. As we describe the commands, try them on your computer. Feel free to explore, as mistakes are never fatal (assuming you remember to save and back up your files—always a good practice!). The chapter will conclude with some guided exercises, reviewing some of the operations outlined here.

The Layout of SPSS

When the SPSS program starts, the program automatically opens to the **SPSS Data Editor Window**. This screen looks and operates like spreadsheet programs. Because there are no data in the SPSS program when it initially starts, the grid is empty and the Data Editor Window says “Untitled.” Your screen should look like Figure 2.1.

SPSS for Windows is menu driven and the program is designed to fulfill common requests easily. There are two types of menus at the top of the window. The **menu bar** contains organized paths to the most commonly requested procedures, such as opening files, performing statistical operations, and constructing graphs. As we work through the SPSS program, you will learn many of the commands in the menu bar. You can explore some of these by drawing the cursor to a command category, such as *File* (Figure 2.2) or *Analyze* (Figure 2.3). To perform any command, hold the mouse button down, move the cursor to the command you wish to perform,

and release the button. SPSS will ask for the specific information it needs to perform that command. This usually includes the variables included in the analysis and the types of output to be reported.

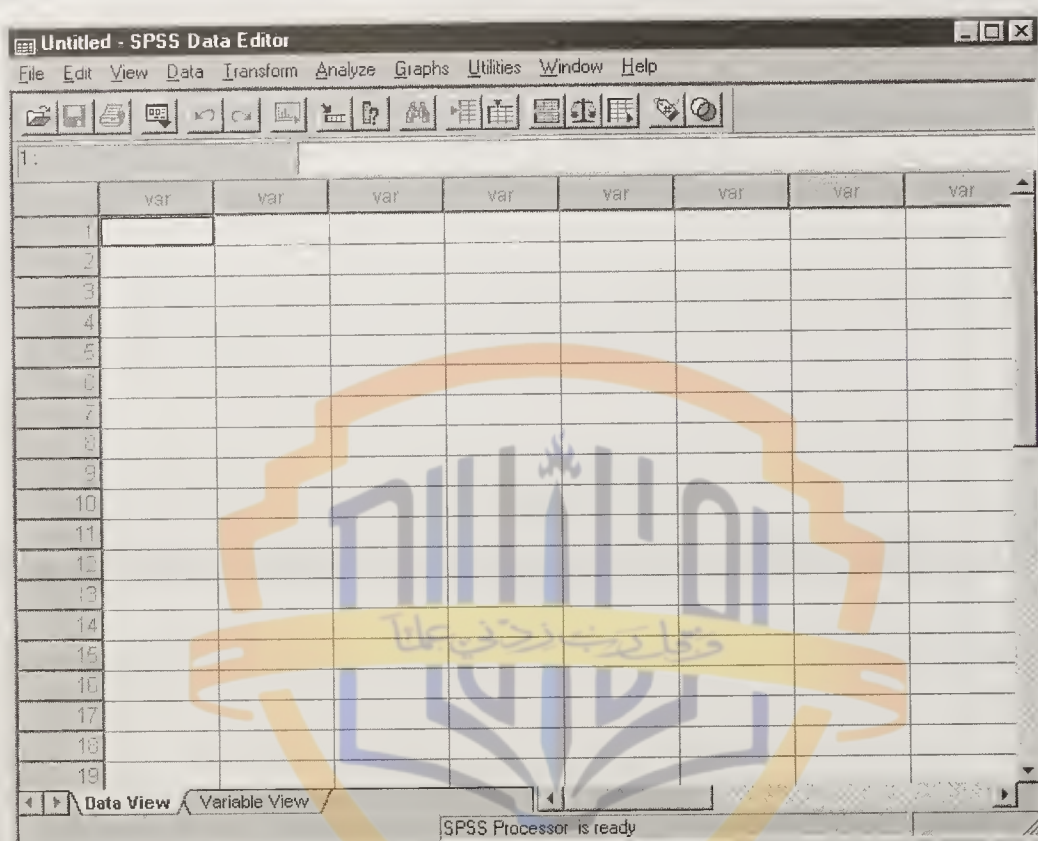


Figure 2.1 The Data Editor Window

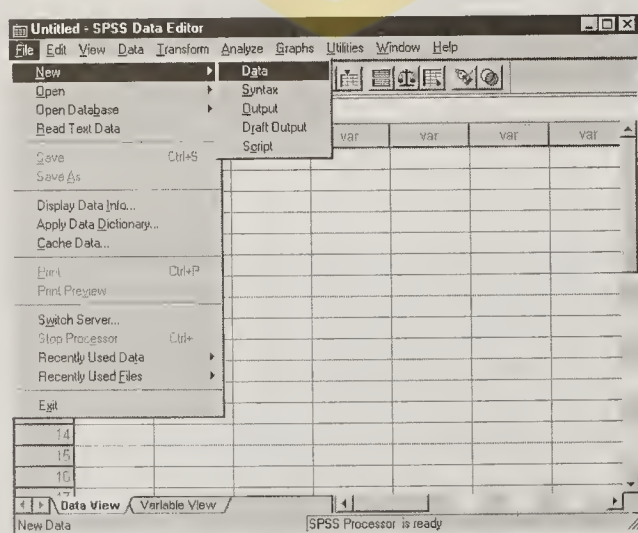


Figure 2.2 File Commands

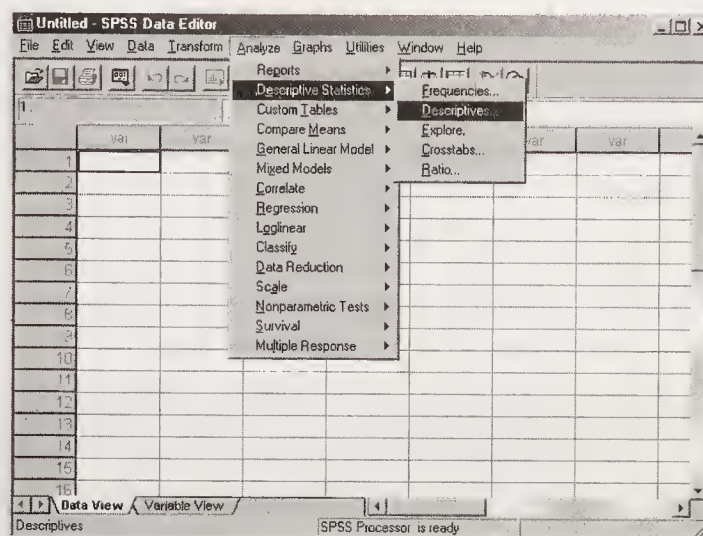


Figure 2.3 Statistics Commands

In this book, we use a shorthand method of describing how to locate specific commands. For instance, if we want you to open a new data file, you will see the following:

File

New

Data

This is a shorthand way of saying: go to the *File* command in the main menu, hold down the mouse button, move the cursor to highlight *New*, and then move the cursor to highlight *Data*. Upon releasing the mouse button, you will have told SPSS to open a new data file and SPSS will give you a window in which you can enter the pertinent information.

Beneath the menu bar is the **tool bar**. This works like the menu and it includes a number of handy devices that refine the analyses. We will not worry about using the tool bar, but as you become more familiar with the program, you will probably find yourself curious about the types of things the tool bar can do. Feel free to experiment.

Not incidentally, there is another way to run commands in SPSS called **syntax commands**. Unlike the menu driven procedures, syntax commands require a greater familiarity with the particular terminology and phrasing necessary for SPSS to perform statistical procedures. It is likely that your professor first learned to use SPSS on earlier versions of the program, which relied exclusively on these syntax commands. For complex and repetitive procedures, this is still the most efficient way to work with SPSS (in fact, this was how we constructed the data sets included in your package). However, for the purposes of the current activities, you will probably be served well by concentrating solely on the menu commands. We will introduce syntax later, as it can be helpful in advanced operations.

Types of Variables

SPSS requires formatting the data so rows in the data editor window represent cases and columns represent variables. Recall from Chapter 1 that **cases** are the units being studied, such as people, organizations, states, or countries. **Variables** are the characteristics being measured for these cases, such as age and gender for people, or per capita income and mortality rates for

states. There are several variable types, each reflecting different ways of assessing and documenting information about each case. One useful way of thinking about variables is to reflect on whether variables indicate categories (such as gender or ethnicity) or numbers (such as age or income). These two classes of variables, categorical and numerical, often require different types of statistical analyses.

Categorical Variables

Categorical variables indicate typologies into which a case might fall. For example, ethnicity (White, Black, Hispanic, Asian) is a categorical variable. Categorical variables can be further distinguished as being either ordinal or nominal. Categories are considered **ordinal** if there is an order to them. For example, a student's class year in school—freshman, sophomore, junior, or senior, is an ordinal categorical variable. Knowing that one student is a senior and another is a sophomore tells us not only that the two students are in different class years, but also that the senior has completed more classes than the sophomore. Categorical variables are considered **nominal** if they do not indicate any order. A student's major is an example of a nominal categorical variable. It categorizes the student and gives information only about what a student is studying, but there is no logical order to psychology, sociology, history, or biology majors. Ethnicity is another example of a nominal variable.

Numerical Variables

Numerical variables give information indicating the quantity of a concept being measured. For example, the number of years a person has spent in school is a numerical variable, as is the number of miles a car can go on a gallon of gas. There are two types of numerical variables: count and continuous. **Count variables** indicate a numerical count and take on the values of whole numbers. For example, the number of classes a student is taking is a count variable. A student can take 3 or 4 classes, but cannot take 3.7 classes.

Continuous variables indicate how much of something exists and can have any value within a range. Continuous variables, therefore, are not restricted to whole numbers. For example, the number of years a student has been in school is a continuous variable. A student could have been in school 13.00 or 13.72 years. The possible values are restricted to a range—the minimum possible value is 0 if a student was just beginning school and the maximum is about 100 (if a very old person was in school his or her entire life). Note, though, that any value between 0 and 100 years is possible, including decimal values. And since decimals are possible, a continuous variable can have an infinite number of possible values (13.72 is a different value than 13.721). Miles per gallon is another example of a continuous variable.

Defining and Saving a New Data Set

With this information on the structure of data sets, we can begin to work with data in the SPSS environment. The first step is to load a data set into the statistical package. There are two ways to do this: manually entering a new data set or opening an already constructed data set.

Data entry immediately follows data collection in a research project. It is also a good way to get a feel for the structure of the SPSS program and how the system operates. In this exercise, we outline the processes involved in constructing a new data set. To illustrate the process, you will generate a data set about your family members. Your data set will contain four variables: Person, Sex, Birth Date, and Education. Each family member will constitute one case

in the data set. In the event that your family is very small, you could also include friends or other relatives.

The data editor contains two spreadsheets – the **Data View** and the **Variable View** spreadsheets. The default spreadsheet is *Data View*, which displays the actual data (Figure 2.4). The rows designate the cases, and the columns designate the variables. The variable names will be at the top of each column. The *Variable View* spreadsheet contains information about each variable (Figure 2.5). Each row of the spreadsheet refers to a variable from your data set and each column contains certain information about that variable.

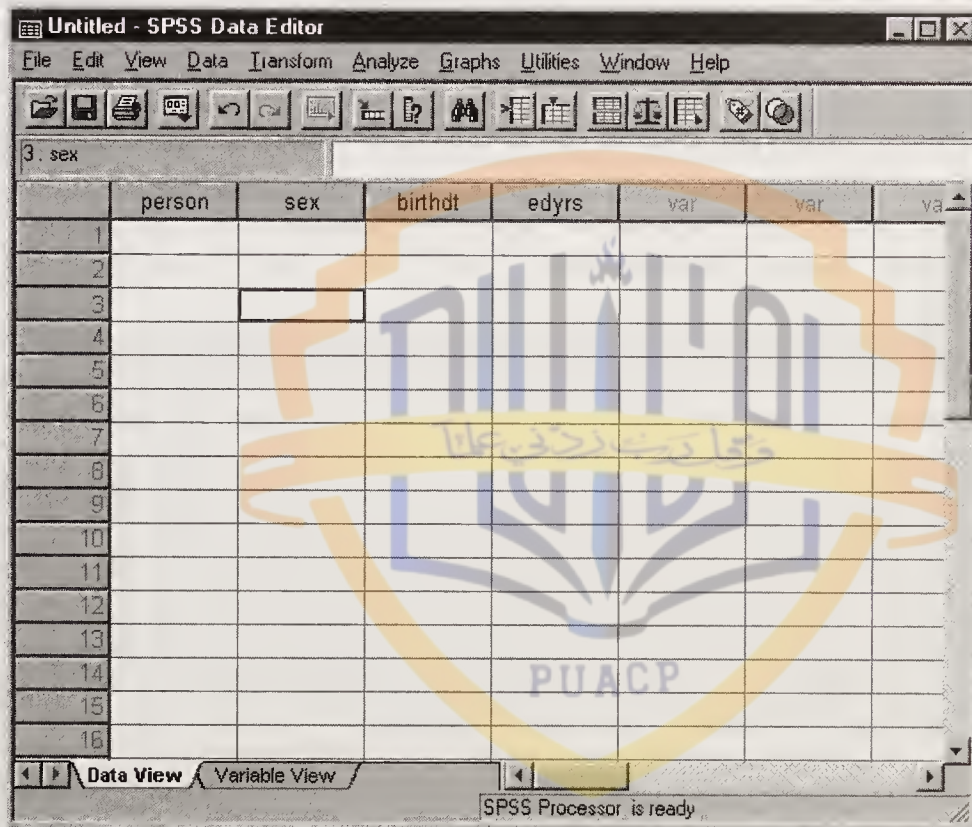


Figure 2.4 The Data View Spreadsheet

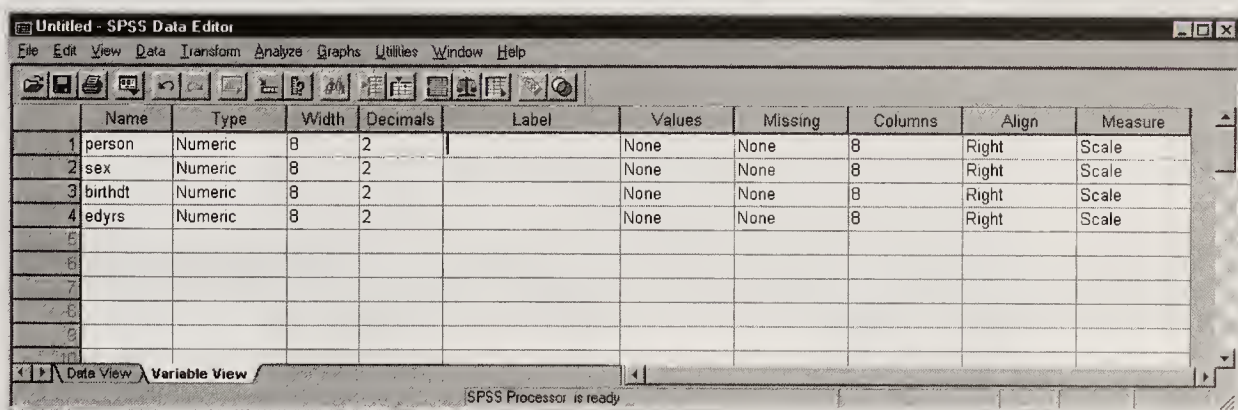


Figure 2.5 The Variable View Spreadsheet

To define the variables in your data set, enter the *Variable View* spreadsheet. There are two ways to open this spreadsheet. The first is to double click the cursor onto the top of any column of the *Data Editor* window. The second way (and the way to get back to the *Data View* when in *Variable View*) is to click on the *Variable View* tab in the lower left corner of the screen.

Once you are in *Variable View*, you can start defining variables. In the column labeled *Name*, enter “Person”. A variable name is restricted to 8 letters, must contain no spaces or symbols, and must start with a letter. Press the enter key or click on the next box to move to the next column. You will notice that many boxes in the rest of the row fill in automatically with default values, some of which we will change.

The next column is *Type*, with the default “Numeric.” Since we want to enter text for people’s names, not numbers, we have to change the default type. If we do not, SPSS will only let us enter numbers for this variable back in the *Data View* spreadsheet. To change the default, click on the small gray box that appears next to “Numeric.” A window will pop open, in which you can choose the type of variable you would like to enter. Since we want to enter text, choose “String” and enter “20” in the box for *Characters*. This informs SPSS that this variable will contain strings of non-numeric information up to 20 characters long and that the data can be treated as text rather than as numerical data. Click “OK” when you are done and notice that SPSS automatically put 20 in the *Width* column.

In the column labeled *Labels*, enter “Family Member Name” as the *Variable Label* (Figure 2.6). The *Variable Labels* help researchers remember what that abbreviated variable name really means. You can skip the rest of the columns for now. Go back to the *Data View* spreadsheet and you will see that the *Data View* now has a new variable in the first column called “Person.” In this column, you can enter data for the cases in your data set, the names of your relatives. Place the cursor in the box intersecting the first column and the first row and click to highlight that box. Enter your own name, and press “Enter.” In the rows below, enter your spouse’s, parents’, siblings’, and children’s names. Your *Data View* spreadsheet should look similar to Figure 2.7. If names are clipped off, the column can be expanded by placing the cursor at the intersection of this column with the next column until the double arrow appears; click and hold the mouse and draw the column to the right.

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure
1	person	Numeric	8	2	Family member's name	None	None	8	Right	Scale
2	sex	Numeric	8	2		None	None	8	Right	Scale
3	birthdt	Numeric	8	2		None	None	8	Right	Scale
4	edyrs	Numeric	8	2		None	None	8	Right	Scale
5										
6										
7										

Figure 2.6 The Variable View Spreadsheet with Variable Label Entered

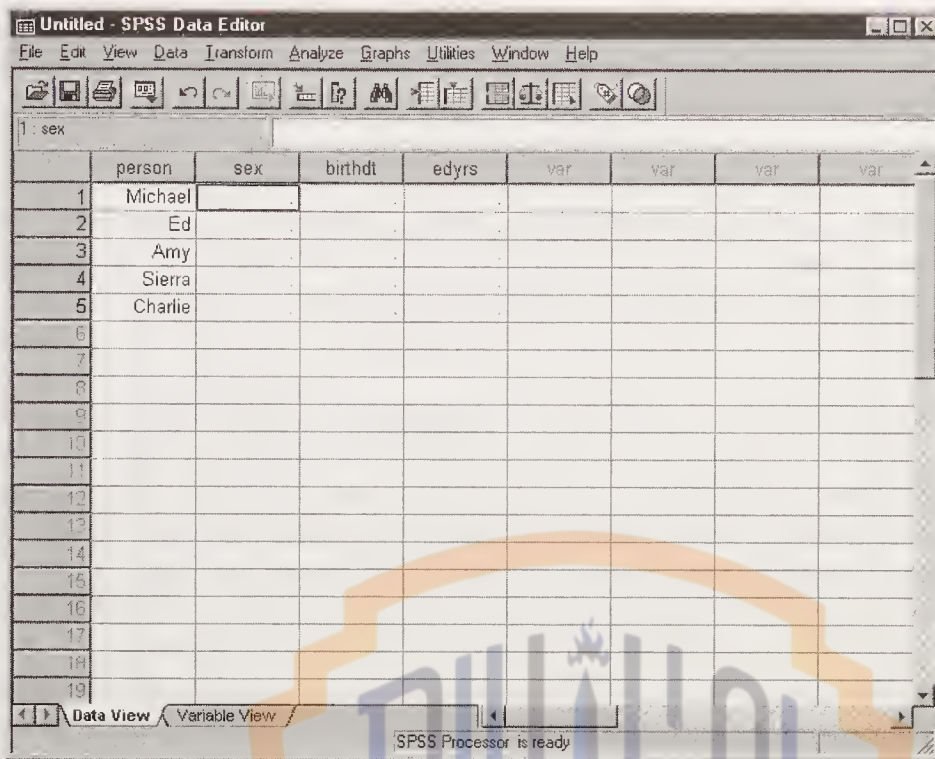


Figure 2.7 Entry of Names with the Data Editor

The other variables can be entered in a similar manner. Return to the *Variable View* spreadsheet, define the next variable as “SEX,” and label it as “Sex of Family Member.” We could make SEX a string variable, but we will keep it a numeric variable with a length of 8 characters and 2 decimal places. Represent males with the number “1” and females with the number “2.” Since it would be very easy to forget which sex we defined as “1” and “2,” it is extremely important to input **value labels**. As a word of warning, it is surprisingly easy to forget seemingly intuitive value labels. It takes a little more time now, but you should always enter both *Variable Labels* and *Value Labels* when defining variables.

Click on the small gray box that appears in the cell in the *Values* column and the *Value Labels* window will pop open (Figure 2.8). Within this window, enter “1” as the *Value* and “Male” as the *Value Label*, and click on *Add*. Do the same for women by entering “2” as the *Value* and “Female” as the *Value Label*. Click on *Add*. Then click on *OK*.

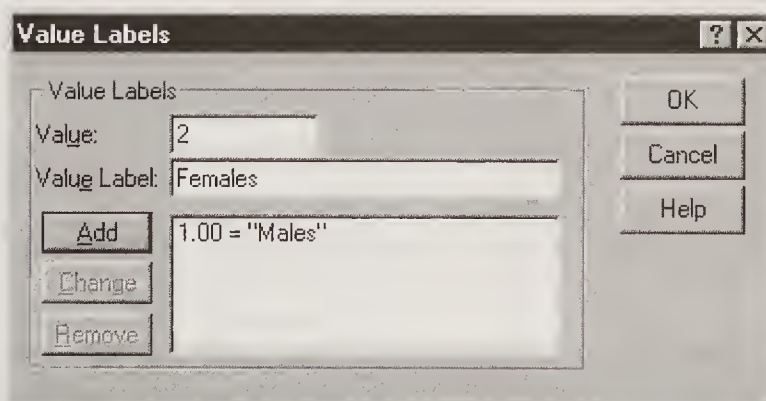


Figure 2.8 Value Labels

The column labeled *Measure* indicates the type of variable that we discussed earlier in this chapter. The default value is “Scale,” which refers to a numerical variable. Notice that for the variable PERSON that you entered in the first row, the default was “Nominal.” This is because PERSON is a string variable, which, by definition, is not numerical. Even if a Nominal variable has numeral values, SPSS will not treat them as having any numerical values, but as letters of the alphabet. Since we left SEX as numerical, SPSS assumes it is a Scale variable. But the numbers 1 and 2 that we used for SEX are merely abbreviations for the categories “Male” and “Female,” so you should indicate to SPSS that SEX is also a nominal variable. If you click on the down arrow next to “Scale,” you have choices of “Nominal” or “Ordinal.” Choose “Nominal.”

Define a new variable for birth date as “BIRTHDT” and *label* the variable “Birth Date.” Because we want this to be treated as a date, click on the small box next to “Numeric” under *Type* and select “Date.” *Date* offers a number of formats for coding the information—select mm/dd/yy. Since dates are numerical (scale) variables, we do not have to change the default *Measure*. Input the appropriate birth dates in the *Data Editor* window.

Highlight the fourth column in the *Data Editor* window. Define a new variable for education as “EDYRS” and *label* the variable “Years of Education.” Because SPSS automatically codes this variable as numeric and scale, there is no need to do any further commands. Your variable view should look like Figure 2.9.

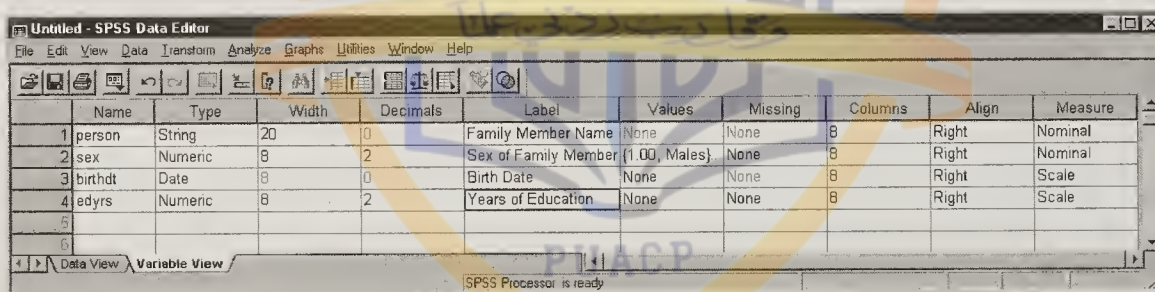


Figure 2.9 The Variable View Window

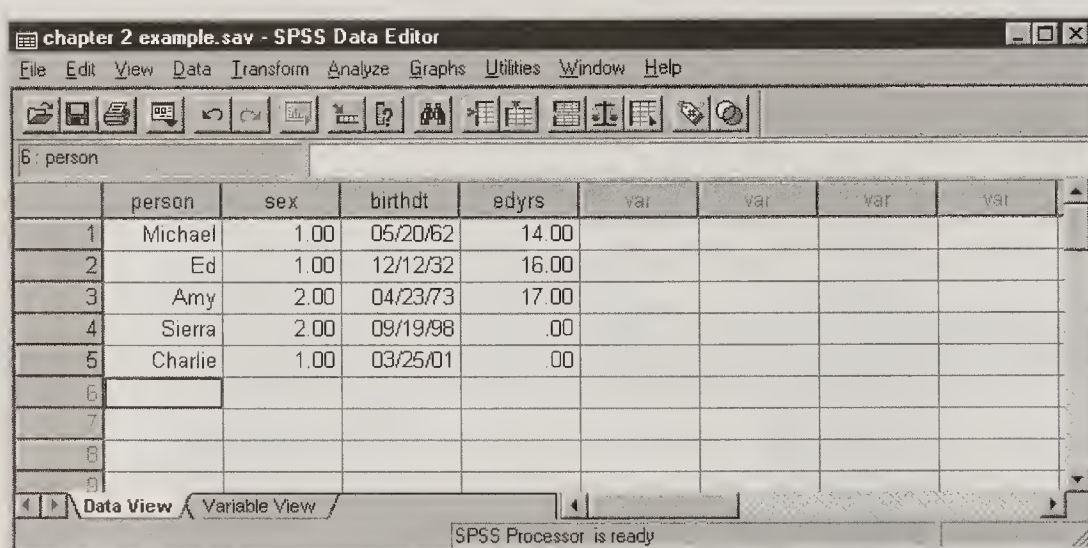


Figure 2.10 The Data View Window

You can now enter the appropriate values for each of your variables in the *Data View* spreadsheet. When you finish, your data set should look like Figure 2.10.

In some circumstances, researchers lack the necessary information to enter a value for a particular case. For example, if you did not know your father's birth date, you should code that datum as "missing." Do this by entering a period (.) as the value.

We will work with the data you compiled here again in the future. For now, however, save your data using:

File

Save As

File Name: FAMILY.SAV

Save

Save the data file on a disk with the name "FAMILY.SAV" We would suggest saving these data on a new disk. Congratulations, you have just defined variables, labeled variables, labeled values, and saved your data.

Managing Data Sets: Dropping and Adding Variables

Good data analysts become skilled at selecting data that are pertinent to their research questions. There is currently so much data available that having too much can become overwhelming. Suppose that you are interested in looking at environmental issues and are compiling a new data set from many other existing data sources. As you compile data concerning recycling, car-pooling, etc., you come across homicide statistics. There might be a relationship between homicide and recycling, but you have to ask yourself at this point, does this new variable have anything to do with my main research question? If it does, keep it in your data set. If it does not, there is no reason to keep it in the data because it will only take up disk space and could later cloud your thinking about what needs to be analyzed.

There are a number of ways to drop variables from the data. One way is simply to go to the top of the *Data Editor* window, highlight the column of the variable that is not desired, and use the *Edit* command:

Edit

Cut

You can try this function by opening your data FAMILY.SAV and cutting the variable EDYRS. You will see this variable disappear from the *Data Editor* window. This variable can be restored by using:

Edit

Undo

It is also possible to add a new variable to the data set using the command:

Data

Insert Variable

Of course, once you insert a new variable, you will need to define it and enter the appropriate values in the same manner described earlier in this chapter. It is important to note that any changes in the data (such as adding or deleting variables) will be saved only if you save the data set before exiting the program. If you do not save the data, any changes will be lost. If you make substantial changes to a data set, it is prudent to save that file under a new name. As we work with data, we usually give the initial file a name such as "FAMILY.SAV." As we refine the data, adding or deleting variables, we save the new files with successive new names such as "FAMILY2.SAV," "FAMILY3.SAV," etc. One advantage of doing this is that it opens the possibility of backtracking and correcting any (inevitable) mistakes made along the way. You might develop a different system, but it is important to keep a running record of changes to your data. It is also important to never copy over the original data set unless you are absolutely sure that you want all of the changes to remain. Once a data set is copied over, the information is changed forever.

Merging and Importing Files

In some circumstances, the data you want to work with may not be in SPSS format. All of the data provided with this text are in SPSS format, which makes your work much easier. However, at some point in your career, you may find it necessary to **import** statistical information from another format, such as Lotus, Excel, dBase, or ASCII. You will find SPSS has the capability of importing an Excel file, for example, with the command:

File

Open

Data

File name: Filename.xls

Files of Type: Excel (*.xls)

Open

In compiling the data for your analysis, we accessed a number of different sources of data, and selected those variables that we thought would be most pertinent for the types of research projects students would be interested in, and then combined these different sources through a process called **merging**.

Data

Merge Files

Add Variables

The above command operates by combining two data sets in such a way that new variables are added on to the end of a primary file. Suppose you had a file containing your family members' social security numbers and wanted to add this information to the data set you compiled earlier. The *Add Variables* command would enable you to do this operation by merging the files according to the selected variables. You would need to first sort both files so that each family member occurs in the same order in each file using the command:

Data

Sort Cases

Once each file is ordered the same, SPSS can match cases so that the appropriate information is added to each case.

It is also possible to add new cases to the data set using the *Merge Files* command. Suppose, for example, that you wanted to combine the information about your family with information about other class members' families. As long as each file has the same variables, new cases (people) can be added using:

Data
 Merge Files
 Add Cases

There is no immediate need for you to import or merge files. However, in the future you may want to add some variables to the data set included with this text or construct an entirely new data set from existing data. If this is the case, you will probably need to use the merge or import procedures. If you have existing data in a format that SPSS cannot import, there are good software packages available, such as DBMS/Copy, that easily translate data files from one format to another.

Loading and Examining an Existing File

As you can tell, entering data can be labor intensive and time consuming. For this reason, many professional social science researchers allot this task to companies that specialize in data entry. Thankfully, social science researchers can often use data already compiled by government agencies or by other researchers. In fact, there are a number of organizations, such as the Inter-university Consortium for Political and Social Research, that specialize in distributing data on the Internet. Your college library probably has government data in computer-ready format as well.

On the disk accompanying this text is a data set containing state level data, STATES.SAV. These data are drawn from a number of sources such as the Department of the Census and the FBI. To examine this data set, use:

File
 Open
 Data
 Look in:
 File Name: STATES.SAV
 Open

Your computer has to search for these data in the correct place, so make sure that *Look in:* (located at the top of the window) directs your machine to search the appropriate drive.

The *Data Editor* window will fill with data from the 50 United States and the District of Columbia. Just as your family members comprised cases in the previous exercise, the states comprise cases in this data set. Your screen should look like Figure 2.11.

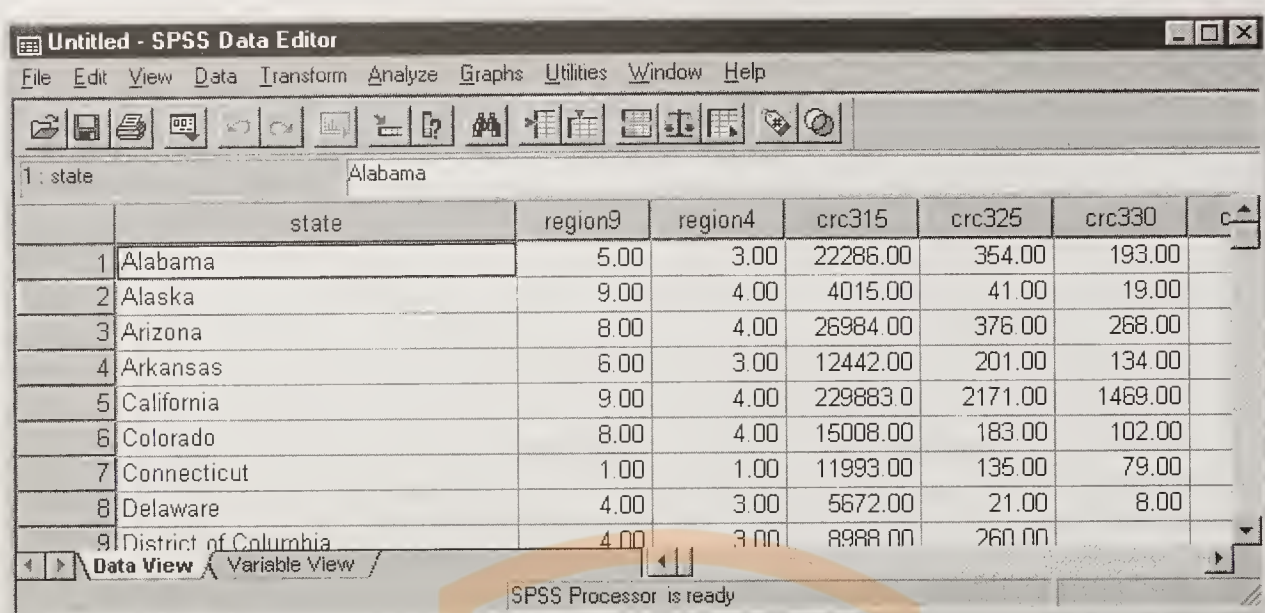


Figure 2.11 shows the SPSS Data Editor window with the STATES.SAV dataset. The 'Data View' tab is selected, displaying a table with 8 columns and 9 rows. The columns are labeled 'state', 'region9', 'region4', 'crc315', 'crc325', 'crc330', and a column with a 'c' icon. The rows represent different US states and the District of Columbia.

	state	region9	region4	crc315	crc325	crc330	c
1	Alabama	5.00	3.00	22286.00	354.00	193.00	
2	Alaska	9.00	4.00	4015.00	41.00	19.00	
3	Arizona	8.00	4.00	26984.00	376.00	268.00	
4	Arkansas	6.00	3.00	12442.00	201.00	134.00	
5	California	9.00	4.00	229883.0	2171.00	1469.00	
6	Colorado	8.00	4.00	15008.00	183.00	102.00	
7	Connecticut	1.00	1.00	11993.00	135.00	79.00	
8	Delaware	4.00	3.00	5672.00	21.00	8.00	
9	District of Columbia	4.00	3.00	8988.00	260.00		

Figure 2.11 The Data Editor Display of STATES.SAV

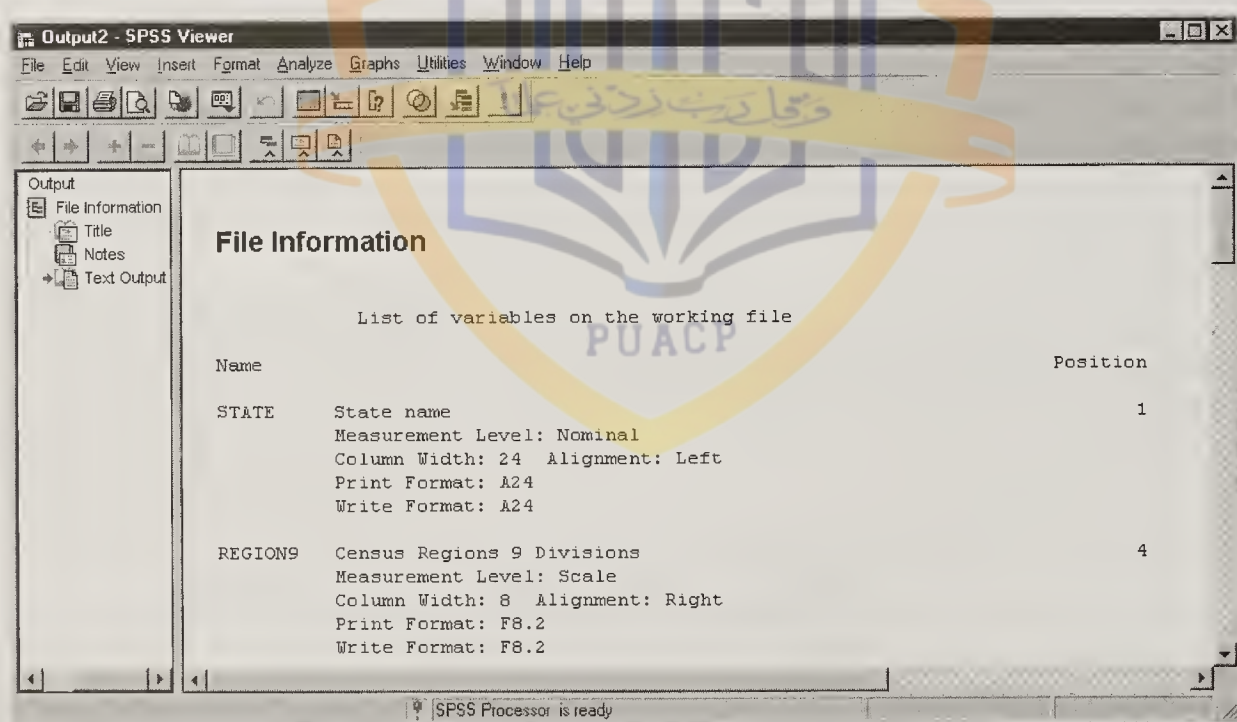


Figure 2.12 shows the SPSS Output Viewer window displaying 'File Information'. The output lists the variables in the working file, their names, measurement levels, column widths, and alignment. The variables listed are STATE and REGION9.

Name	Position
STATE State name Measurement Level: Nominal Column Width: 24 Alignment: Left Print Format: A24 Write Format: A24	1
REGION9 Census Regions 9 Divisions Measurement Level: Scale Column Width: 8 Alignment: Right Print Format: F8.2 Write Format: F8.2	4

Figure 2.12 The Output Window for File Info

There are many variables in this data set, including indicators of population, educational attainment, income, and health care. Each variable is represented by a name, shown across the top row of the *Data Editor* window. One easy way to find the list of all of the variable names and variable labels is to use:

Utilities

File Info

This command will display all of the variable names and labels, as well as yield information on how these variables are defined (e.g., string, numeric, or date) in the **SPSS Output Window**. Your output should look similar to Figure 2.12.

Managing Variable Names and Labels

You may have noticed that the data in the STATES data set have names that appear cumbersome. For example, why not just name variable JCC186 (Reported Juvenile Arrest Rate in 1998) JuvenileArrests? Remember, SPSS limits variable names to 8 characters or fewer. This means that almost all variables will need truncated names. Because there are so many variables relating to juvenile crime in this data set, we decided to indicate this with the initials JC (the first two letters in the name “juvenile crime”). The remaining letters and numbers indicate the source of the data. In the case of the STATES data set, all of the data have been drawn from three books:

Crime State Rankings 2000 (Morgan Quitno 2000)

Health Care State Rankings 2000 (Morgan Quitno 2000)

State Rankings 2000 (Morgan Quitno 2000)

The third letter in the variable name indicates the source of the data. C = *Crime State Rankings 2000*, H = *Health Care State Rankings 2000*, S = *State Rankings 2000*. The remaining numbers indicate the page of the text from which these data were drawn. Using these variable names and these texts, the data can be used to trace the data back to its original sources.

For example, variable JCC186 can be interpreted as:

JC- Juvenile Crime

C- *Crime State Rankings 2000*

186- Page 186.

To simplify searches of the data for specific subjects, below is a list of abbreviations used in the variable names in the STATES data.

Variable Name Prefixes

ar - arrests

bi - births

ca - child abuse

cr - crime

df - defense

di - disease

dm - demography, population

dr - drugs and alcohol

dt - deaths

ec - economy

em - employment

en - environment

fi - finance

hb - health behaviors

jc - juvenile crime

le - law enforcement

pf - physical fitness

ph - health providers

pr - prisons

pv - poverty

sc - schooling

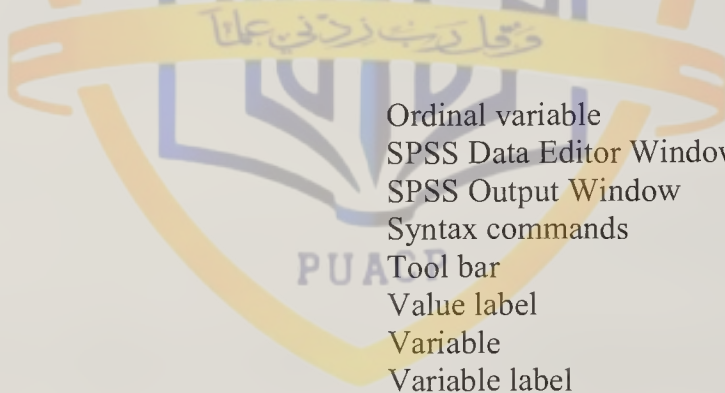
On the surface, this approach to naming variables seems cumbersome. However, it offers a few advantages: it is systematic and allows the researcher to locate the variables quickly; it groups similar concepts together alphabetically; and it enables location of the original source of the data. After gaining experience in working with data, you will develop your own systems for naming and labeling variables.

SPSS also limits how long variable labels can be. Labels can be up to 120 characters long, but often these labels will be cut off in some of the SPSS output. We generally restrict labels to about 80 characters so that they can be fully displayed. However, this often necessitates abbreviating words in complicated or lengthy labels. In the STATES data, we have been forced to abbreviate in a number of circumstances. In most cases, the abbreviation will be apparent. Appendix 3 is a list of common abbreviations.

Summary

SPSS has the capacity to import data from other computer programs, to combine data from different sources into a single data set, and to facilitate the creation of an entirely new data set. In this chapter, you learned how to construct a data set and how to retrieve existing data. We also discussed data management and some of the skills to select appropriate data and develop systematic methods of saving and naming data sets. A well constructed data set is the foundation of all subsequent analysis, so particular attention needs to be paid to designating variable types, labeling variables clearly, and naming variables in a way so they are accessible to users.

Key Terms



Case	Ordinal variable
Categorical variable	SPSS Data Editor Window
Continuous variable	SPSS Output Window
Data view spreadsheet	Syntax commands
Count variable	Tool bar
Import	Value label
Menu bar	Variable
Merging	Variable label
Nominal variable	Variable view
Numerical variable	

References and Further Reading

- Green, Samuel B., Neil J. Salkind, and Theresa M. Akey. 1999. *Using SPSS for Windows: Analyzing and Understanding Data*. Upper Saddle River, New Jersey: Prentice Hall.
- Morgan, Kathleen O'Leary, Scott Morgan, and Neal Quitno. 2000. *State Rankings 2000*. Lawrence, Kansas: Morgan Quitno.
- Morgan, Kathleen O'Leary, Scott Morgan, and Neal Quitno. 2000. *Crime State Rankings 2000*. Lawrence, Kansas: Morgan Quitno.
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Chapter 2 Exercises

Name _____

Date _____

1. Create a new data set by polling 5 other people and asking the following questions:

RESPOND. What are the last four digits of your phone number?

EXER1. Do you exercise regularly? (Yes or No)

EXER2. How many hours a week would you say you exercise?

EXER3. Do you participate in team sports? (Yes or No)

EXER4. What is your favorite sport to play?

Create a new data set using these data and incorporating the following labels.

Variable Name	Label	Value Labels	Variable Type
RESPOND	Respondent Number	None	Numeric
EXER1	Exercise Regularly?	0=no 1=yes	Numeric
EXER2	Hours/Week of Exercise	None	Numeric
EXER3	Participate in Team Sports?	0=no 1=yes	Numeric
EXER4	Favorite Sport	None	String 20

Input the responses and save the data as “EXERCISE.SAV”. Print *File Info* of all of the variables. Print a copy of the Data Editor window.

2. Open the data STATES.SAV. Examine the data set using the *File info* command and locate some variables that measure the following concepts. (Double click within the box of variables and scroll up or down to view all of the variables.)

A. Economic Prosperity

_____ Variable Name	_____ Variable Label
------------------------	-------------------------

_____ Variable Name	_____ Variable Label
------------------------	-------------------------

_____ Variable Name	_____ Variable Label
------------------------	-------------------------

B. Physical Fitness

_____ Variable Name	_____ Variable Label
------------------------	-------------------------

_____ Variable Name	_____ Variable Label
------------------------	-------------------------

_____ Variable Name	_____ Variable Label
------------------------	-------------------------

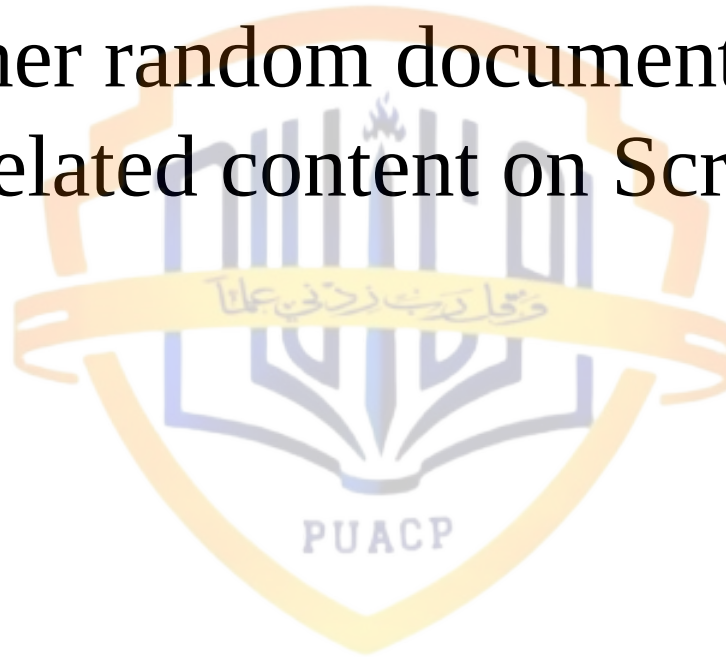
C. Educational Attainment

_____ Variable Name	_____ Variable Label
------------------------	-------------------------

_____ Variable Name	_____ Variable Label
------------------------	-------------------------

_____ Variable Name	_____ Variable Label
------------------------	-------------------------

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